JAMES CHRISTOPHER MITCHELL

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PROFESSIONAL SUMMARY

I'm a Machine Learning Engineer with 12 years of experience building production systems that process billions of data points daily. I've spent most of my career at the intersection of research and engineering, taking academic ideas and turning them into systems that actually work at scale. I enjoy mentoring junior engineers and have led teams ranging from 3 to 8 people. My particular strength is taking on ambiguous problems where the path forward isn't obvious and shipping something that measurably improves business outcomes.

WORK EXPERIENCE

Staff Machine Learning Engineer

Synapse Data Ltd

Remote (Cambridge base)

July 2021 - Present

Designed and deployed a real-time recommendation system serving 2.3 million daily active users with sub-100ms latency requirements, increasing engagement metrics by 34 percent.

Led a cross-functional team of 6 engineers and data scientists to rebuild our feature engineering pipeline, reducing training time from 8 hours to 45 minutes and enabling daily model retrains.

Implemented automated model monitoring and drift detection systems that reduced production incidents related to model degradation by 87 percent over 18 months.

Mentored 4 junior machine learning engineers, with 3 progressing to senior roles or moving into staff-track positions within the organization.

Senior Machine Learning Engineer

DeepMind Technologies Ltd

London, United Kingdom

www.deepmind.com

January 2021 - March 2021

Developed novel approaches to multi-task learning that improved model performance across 12 different prediction tasks simultaneously, achieving state-of-the-art results on internal benchmarks.

Contributed to research publication on uncertainty quantification in neural networks that received 340+ citations within two years of publication.

Optimized model inference pipeline for edge deployment, reducing model size by 68 percent while maintaining 99.2 percent accuracy on validation sets.

Collaborated with product teams to translate research findings into 3 shipping features used by millions of users globally.

Machine Learning Engineer

Barclays Bank PLC

London, United Kingdom

www.barclays.com

September 2012 - December 2020

Built and maintained credit risk models handling over GBP 450 billion in annual transaction volume, with regulatory approval from the FCA and PRA.

Developed anomaly detection system for fraud prevention that reduced false positives by 52 percent while maintaining 99.1 percent detection rate, preventing approximately GBP 18 million in losses annually.

Established reproducible model development practices and introduced MLOps tooling that became the standard for 40+ data scientists across the division.

Managed vendor relationships and evaluated third-party machine learning platforms, ultimately selecting and implementing solutions that reduced infrastructure costs by 31 percent.

EDUCATION

PhD Computer Science (Machine Learning)

University of Cambridge

Cambridge, United Kingdom

2008 - 2012

BSc Natural Sciences (Physics and Mathematics)

University of Edinburgh

Edinburgh, United Kingdom

2004 - 2008

TECHNICAL SKILLS

Machine Learning: Deep learning, reinforcement learning, time series forecasting, anomaly detection, recommendation systems, transfer learning, multi-task learning

Programming Languages: Python, Scala, Java, SQL, C++

Data & ML Frameworks: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Spark MLlib

Data Engineering: Apache Spark, Airflow, Kafka, BigQuery, PostgreSQL, Elasticsearch

Model Deployment: Docker, Kubernetes, AWS SageMaker, model serving, A/B testing frameworks

Data Visualization: Matplotlib, Plotly, Tableau